

MathGov 3R / 1-2 Two-Phase Public Introduction v12.4

v12.4 Current Release Integration

Release: MathGov Core Release 2026.09 v12.4 / SGP v8.3 - Calculability, Type-Integrity, and Cross-Artifact Synchronization Public Research Source Release

Exact current companion pins: Canon v12.4; SGP v8.3; ripple.md v5.3; Agent System v12.2; CSV v2.2; Cascade v2.4; Reproducibility v1.2; WDBIP v1.4; RLS Validation v2.4; Primer v4.2; Public Introduction v12.4; PC-AEP/MFDI/Source-Coupling v2.1; Aligners Sheet v5.4.

Integrity surface	Current v12.4 requirement	Claim boundary
3R	Reality. Rights. Ripples.	A public doorway, not the complete technical specification.
1-2	Qualify through RG, RF/NCRC, TRC, and CSV; then rank selectable options through RLS.	Ranking cannot rescue prior failure.
Eligible shorthand	Eligible(a) is teaching shorthand for survival of the declared qualification sequence.	It is not a sixth gate, new authority predicate, or execution licence.

This is the public doorway, not the full specification. Use it to understand the rhythm; use the Canon for conformance. Informative, non-normative public teaching companion. Canon controls if any conflict occurs.

Release: MathGov Core Release 2026.09 v12.4 / SGP v8.3 - Calculability, Type-Integrity, and Cross-Artifact Synchronization Public Research Source Release.

v12.4 Patch Note

v12.4 corrects the current public release identity and claim boundary. The 3R and MathGov 1-2 teaching structure is unchanged.

Use the smallest profile that is safe

- **Quick / Tier 1:** a short record for low-stakes, reversible choices.
- **Standard / Tier 2:** evidence, parameters, gates, controls, and uncertainty for material organizational or community choices.
- **Audit / Tier 3:** a full reproducible record, hashes, independent review, and replay for high-stakes or public decisions.

The framework stays the same at every level. Only the documentation burden changes. A user must escalate when rights, ruin, vulnerable groups, physical execution, consequential AI authority, or public validation claims become material. See

[docs/implementation/MATHGOV_REPRODUCIBILITY_AND_USE_STANDARD_v1.2.md](#).

The shortest version

Reality. Rights. Ripples.

1. Qualify. Reality. Rights. Ruin. Viability.

2. Rank. Ripple-score the viable options.

That is the operating rhythm.

First make sure the option belongs in the ring. Then find the one that produces the best ripple.

The full formal cascade

RG -> RF -> TRC -> CSV -> RLS

Reality Grounding -> Rights Floor -> Tail-Risk Constraint -> Containment and Structural Viability -> RippleLogic Scoring

Teaching note: this doorway compresses the architecture into three pillars and two phases. The formal architecture has five levels: one grounding precondition, three qualification gates, and one ranking layer.

Why this matters

MathGov does not begin by scoring every imagined option. It first asks which options have enough reality, rights protection, ruin control, and viability to enter the decision field. Only then does RippleLogic score the survivors.

The MathGov answer is: ground the claim, protect the right, bound the ruin, verify the system, make exit and correction real, and rank only what survives qualification.

The three pillars

Pillar	Plain meaning	Formal location
Reality	Do we have contact with what is real, possible, grounded, testable, bounded, auditable, correctable, and viable?	RG opens reality contact. CSV tests operational reality and structural viability. PC-AEP is used when physical or causal execution claims matter.
Rights	Are protected beings, dignity, consent, thresholds, and non-collapse safeguards protected?	RF/NCRC protects rights directly. TRC protects against ruin pathways that can destroy the conditions for rights and flourishing.
Ripples	Among the options that survive qualification, which creates the best net flourishing?	RLS ranks only the survivors across affected unions and welfare dimensions.

v12.2 rights and ruin hardening

Rights hardening: RF/NCRC does not discount a categorical or severe rights violation merely because it is brief or evidence confidence is low. Rights severity, categorical prohibitions, and severe-hazard exposure are checked separately from ordinary welfare timing.

Tail Emergency Mode: if every rights-safe option exceeds the ruin corridor, the default is redesign, delay, escalation, no-action review, or refusal. A temporary least-CVaR path is allowed only under a governed emergency declaration, absolute exposure cap, independent challenge, time bounds, controls, and exit criteria. It is not ordinary selection, and RLS is not used to rescue it.

Phase 1: Qualify the options

Phase 1 asks: which options are even eligible to be ranked?

Gate	Question	If the answer fails
RG - Reality Grounding	Is the option grounded in evidence, real constraints, affected unions, and a clear claim boundary? Can high-consequence claims be tested or warranted, bounded, audited, corrected, and refused?	Do not score fantasy. Narrow, investigate, mark exploratory, escalate, or refuse the stronger claim.
RF - Rights Floor	Does the option protect rights, dignity, consent, and non-negotiable safeguards?	Do not average rights away. Redesign, escalate, or refuse.
TRC - Tail-Risk Constraint	Is catastrophic, irreversible, fat-tail, or system-breaking risk bounded?	Do not gamble ruin for ordinary benefit. Bound, redesign, escalate, or refuse.
CSV - Containment and Structural Viability	Can the option actually stand, execute, be controlled, monitored, exited, corrected, and remain governable?	Do not select what cannot stand. Add controls, narrow, redesign, or refuse.

TRC is the ruin veto. CSV is the viability and control test. They are linked, but they should not be collapsed.

Reality Grounding hardening

For life-critical systems, weapons, nuclear command and control, planetary-scale cognition, critical infrastructure, medical systems, ecological interventions, or other high-consequence domains, no claim should govern action unless it can be tested or otherwise warranted, bounded to a validity domain, audited with traceable evidence, corrected through monitoring and revision, and refused when the evidence surface is insufficient.

In public language: ground the claim; protect the right; bound the ruin; verify the system; make exit and correction real; rank only what survives qualification.

Phase 2: Rank the survivors

Phase 2 asks: among the options that survived qualification, which one creates the best ripple?

RLS evaluates affected unions, welfare dimensions, evidence, uncertainty, robustness, and decision state.

Mathematical spine

Wrong method:

Decision = $\text{argmax RLS}(a)$ over all imagined options

Correct method:

$\text{Eligible}(a) = \text{RG_qualified}(a) \text{ AND } \text{RF/NCRC}(a) \text{ AND } \text{TRC_qualified}(a) \text{ AND } \text{CSV_qualified}(a)$

Here RG_qualified means RG_SUPPORTED or RG_NARROWED; TRC_qualified means TRC_PASS or a documented TRC_NOT_TRIGGERED assessment. Eligible remains teaching shorthand, not a machine-state token, sixth gate, or execution licence.

Then:

$A_{\text{sel}} = \{a : \text{RG_qualified}(a) \text{ AND } \text{RF/NCRC}(a) \text{ AND } \text{TRC_qualified}(a) \text{ AND } \text{CSV_qualified}(a)\}$

Then apply the decision-state rule:

- If A_sel is empty, redesign, delay, escalate, use the governed emergency pathway where applicable, or refuse the stronger claim.
- If RLS is defined and produces one unique decisive leader in A_sel, the framework may return ALLOW_FRAMEWORK_SELECTION.
- If RLS is undefined, tied, close, uncertainty-overlapped, or non-decisive, return REFUSE_DETERMINISTIC_SELECTION. Seek more evidence, preserve multiple options, redesign, or use a separately disclosed AuthoritySelectionRecord within a lawful mandate.

A point-score argmax is therefore a comparison aid inside the selectable set. It is not automatic authorization, and it does not erase uncertainty, ties, refusal, or the separation between framework selection and authority selection.

No RLS rescue

A high RippleLogic score cannot rescue failed reality grounding, a rights violation, an unacceptable ruin path, an unsupported physical-execution claim, or a structurally non-viable option.

Decision outcomes

Outcome	Meaning
SELECT	The option is eligible and clearly best among survivors.
SELECT WITH CONTROLS	The option is eligible only with safeguards, monitoring, or bounds.
NARROW	The claim, scope, domain, or time horizon must be reduced.
REDESIGN	The option fails but may be improved.
ESCALATE	Authority, evidence, expertise, or stakeholder participation is insufficient.
REFUSE	The requested claim, selection, or action is not supportable under the current evidence, authority, or gate state. Record the specific reason.
NON-DECISIVE	The evidence or scores do not justify deterministic selection.

Public script

MathGov is a decision architecture, not a claim that every option should be scored. It first qualifies options through reality, rights, ruin, and viability. Then it ranks only the surviving options by their ripple effects across affected unions. In short: qualify before you rank.

The 60-second checklist

1. Qualify

- Is it real?
- Are rights protected?
- Is ruin bounded?
- Can it stand, be monitored, be corrected, and be exited?

2. Rank

- Who and what is affected?

- Which option creates the best ripple among the survivors?
- Is the result robust enough to select, or should it be non-decisive?

Claim boundary

MathGov Core v12.4 is a proposed Tier 1-3 auditable decision architecture. It is not empirical validation, legal certification, deployment certification, Tier 4 ProofPack, automated moral truth, or physical safety proof. For consequence-bearing physical systems, MathGov requires external domain evidence through PC-AEP; it does not manufacture that evidence by itself.

Final pocket form

Reality. Rights. Ripples.
Qualify before you rank.
Rank only what survives.

SGP v8.3: protection, participation, and power are separate

When a decision may affect a welfare-bearing being, the current SGP interface does not ask one number to answer every question.

- **MPS** asks how strongly evidence supports welfare-bearing experience.
- **FPP** records full non-downgradable rights-of-protection.
- **GPR** asks how the entity can participate directly, with support, or through representation.
- **SPR** asks whether it may hold one specifically named consequential role under strict gates.
- **ICP** describes intelligence and understanding, but cannot establish sentience or authority by itself.
- **RMCP profiles integrated reality-management capacity; P100 is its open top plateau. Every human has FPP-100, while humanity collectively is the provisional current RMCP-P100 reference anchor. Other biological, digital, hybrid, collective, or extraterrestrial intelligences may reach or already occupy P100 under admissible evidence.**

RMCP/P100 is not a moral-worth or authority score. The operating rule is: **unknown is not zero; capability is not welfare; protection is not authority; full protection does not require human likeness.** MPS intervals may inform residual welfare sensitivity only after rights, ruin, and viability qualification.

SGP interface clarification. MPS is not a cardinal welfare multiplier or moral-worth score. It determines protection posture and which welfare-inclusion hypotheses must be tested. Missing evidence is MPS-NE, not zero.